

Application of Multivariate Data Analysis for Identification and Successful Resolution of a Root Cause for a Bioprocessing Application

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Multivariate Data Analysis (MVDA) can be used for supporting key activities required for successful bioprocessing. These activities include process characterization, process scale-up, process monitoring, fault diagnosis and root cause analysis. This paper examines an application of MVDA towards root cause analysis for identifying scale-up differences and parameter interactions that adversely impact cell culture process performance. Multivariate data analysis and modeling were performed using data from small-scale (2 L), pilot-scale (2,000 L) and commercial-scale (15,000 L) batches. The input parameters examined included bioreactor pCO₂, glucose, lactate, ammonium, raw materials and seed inocula. The output parameters included product attributes, product titer, viable cell density, cell viability and osmolality. Time course performance variables (daily, initial, peak and end point) were also evaluated. Application of MVDA as a diagnostic tool was successful in identifying the root cause and designing experimental conditions to demonstrate and correct it. Process parameters and their interactions that adversely impact cell culture performance and product attributes were successfully identified. MVDA was successfully used as an effective tool for collating process knowledge and increasing process understanding.

Introduction

The increasing use of multivariate data analysis (MVDA) in both basic research and applied scientific fields has enabled the diagnostic evaluation of parameter interactions that were previously undefined. Data sets originating from the manufacturing of biopharmaceuticals are complex; therefore univariate or bivariate analysis is often inefficient resulting in misleading conclusions (Kourti, 2004). Key information in such cases lies in the correlation structure between variables and can lead to spurious results when tested independently. Multivariate data analysis by means of projection methods overcomes challenges associated with such applications, such as multidimensionality of the data set, multicollinearity, missing data and variation introduced by deviating factors such as experimental error and noise (Martin et al., 2002). Principal Component Analysis (PCA), Partial Least-Squares (PLS), and multiple regression are some of the commonly used projection methods. These methods can project process data on lower dimensional spaces for easy inspection (Kourti, 2004). PCA is used to reduce multidimensional data sets to lower dimensions for analysis. It summarizes the information in the observations as a few new variables that are called principal components. It then derives a model (finds lines, planes in the K-dimensional variables space) that fits the data maximizing the variance of the projection coordinates (Wold, 1987). Residual variance (noise) is minimized and variance of the scores (coordinates of the line) is maximized. PLS is a multivariate statistical method that analyses the covariances between the sets of variables rather than optimizing linear combinations of variables in the various sets (Wold, 2001). It is a projection of **X** variables matrix guided by **Y** response matrix. PLS is a compromise between PCA (optimal projection

of **X** matrix) and multiple regression (maximize relationship to **Y** response). It is a dual technique that tries to find directions in **X** that both characterize the **X** matrix well and are related to **Y**.

More recently, MVDA has been applied to various applications in the biopharmaceutical industry (Gabrielsson et al., 2002; Johnson et al., 2007). Applications include near-infrared spectral information from an antibiotic production process (Vaidyanathan et al., 2001), nonlinear multivariate analysis and artificial autoassociative neural network (AANN) for bioprocess fault detection (Shimizu et al., 1998), multivariate statistical process monitoring for processing of pharmaceutical granules (Ündey and Çinar, 2002), the assessment of seed inoculum quality from a manufacturing process (Cunha et al., 2002) and development of an integrated online multivariate statistical process monitoring, product attributes prediction and fault diagnosis framework for a fed-batch penicillin fermentation (Ündey et al., 2003).

Gunther et al. have applied a flexible process monitoring method to pilot plant cell culture data for fault detection and diagnosis (Gunther et al., 2007). A principal component analysis (PCA) model was constructed from 19 batches and the model was shown to successfully detect abnormal process conditions and diagnose root causes. Kirdar et al. examined the feasibility of using multivariate data analysis (MVDA) for supporting key activities required for successful manufacturing of biopharmaceutical products, including scale-up, process comparability, process characterization and fault diagnosis (Kirdar et al., 2007). Multivariate data analysis and modeling were performed using representative data from small-scale (2 L) and large-scale (2,000 L) cell-culture batches. Several input parameters (pCO₂, pO₂, glucose, pH, lactate and ammonium ions) and output parameters (purity, viable cell density, viability, osmolality) were evaluated in this analysis. Scores plots, Loadings plots and

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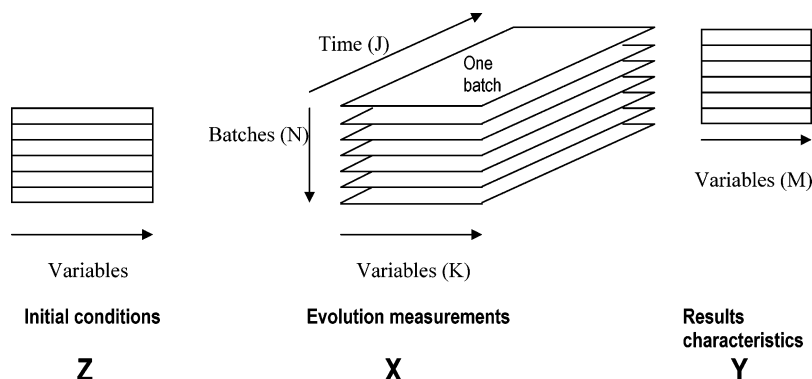


Figure 1. Illustration of batch process data.

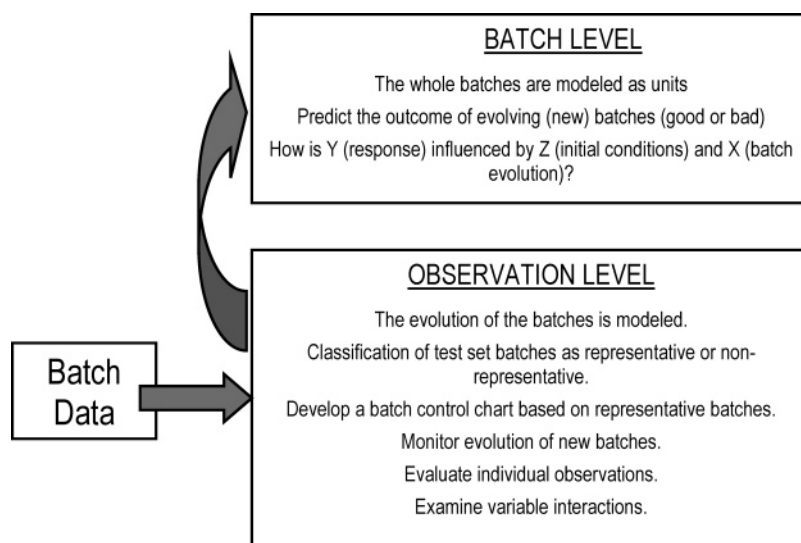


Figure 2. Overview of BSPP via batch modeling.

Variable Importance for the Projection (VIP) plots were utilized for assessing scale-up and comparability of the cell-culture process. Batch control charts were also shown to be useful for fault diagnosis during routine manufacturing. Further, observations from reviewing VIP plots were in agreement with conclusions from process characterization studies demonstrating the effectiveness of MVDA as a tool for extracting process knowledge.

In this paper, MVDA was performed on data from small-scale, pilot-scale and commercial-scale runs. Process parameters and their interactions that adversely impact cell culture performance and product attributes were successfully identified. It has been demonstrated that MVDA is an effective diagnostic tool for identification of root cause and designing experimental conditions to demonstrate the hypothesis and propose a solution.

Materials and Methods

Software. A commercially available MVDA software package, SIMCA P+ 11 version 11.0.0.0 (Umetrics AB, Kinnelon, NJ), was used to perform the multivariate analysis. Prior to analysis in SIMCA, process data were assembled in Excel (Microsoft, Redmond, WA).

Multivariate Data Analysis (MVDA) Method. Batch Modeling. As seen in Figure 1, batch processes yield a matrix, X , which can be illustrated as a three-dimensional data table composed of data collected for each process variable (K) over

a defined time interval (J) for a number of batches (N). A number (M) of final results (such as product titer and product attributes) are designated in the data table Y .

When using partial least-squares (PLS) as the projection method, a space with K and M dimensions is created for each matrix X and Y . Every observation in a data set can be visualized as one point in X -space and another point in the Y -space. Thus, hyperplanes called principal components are calculated to provide maximum correlation between points in the X - and Y -spaces.

Each observation is then projected onto this hyperplane and translated into latent variables known as scores (t for X -space projections, u for Y -space projections). Based on these score values, weights are assigned to express the correlation between points in the X - and Y -spaces, respectively. Weights (w for X -space projections, c for Y -space projections) are assigned on the basis of the variable's influence on the model at any given point in the batch evolution.

The quality of a batch depends on the evolution of the batch and the initial conditions of the batch, such as raw materials. Using historical data of acceptable (good) batches, models of batch evolution and batch completion can be generated creating a summary of all the variables measured. The monitoring of batches in real time may allow for corrective action. The approach used for batch statistical process control (BSPP) is illustrated in Figure 2. This approach uses all available batch data and analyzes data on two levels, observation level

and batch level. These two linked monitoring spaces were used: the observation level to monitor the evolution and the batch level to monitor completed batches. Using MVDA of batch data enables the correlation structure among measured variables to be investigated, while separating representative (good) batches from non-representative (poor) ones. By understanding the properties that dominate a process, comparing batch-to-batch variations, real-time process monitoring and early fault detection are also possible. Both the observation level and batch level modeling approaches are used for detection of abnormal batch operation. The observation level, however, is particularly useful for use in real-time applications for fault detection early in the batch trajectory. The batch level approach is useful for historical analysis of batch to batch comparisons and for modeling interactions between initial conditions, the process evolution and final conditions.

Observation Level Modeling. In order to accomplish observation level modeling, the three-way batch data table (Figure 1) is unfolded, preserving variable direction to a two-way data table. This type of unfolding has previously been shown to be successful for detection of abnormal batches (Wold, 1987, 1998, 2001 and Ündey et al., 2003). As described above, modeling was performed by projecting observations on the hyperplanes and translating them into latent variables. PLS was used to relate the process data to a maturity-related $\mathbf{Y}$ -variable representing relative local batch time. This provides an appropriate maturity index model that can be used to explore how far a batch has progressed.

Next, a model diagnostics step is performed to check that there are no outliers among the reference batches. Outlier batches are excluded if a clear abnormality is found, and the model building process is then repeated. Finally, the batch prediction control charts (scores giving the trajectory of a batch) are created to differentiate between representative and non-representative test set batches (Kirdar et al., 2007).

Batch Level Modeling. At the batch level, all available data are used to model the whole batches as units. In contrast to the previous level, where each row corresponded to one time point in one particular batch, each row in a data table at the batch level now represents one whole batch. In creating a batch level model you can choose to use the original variables or the scores derived from the lower observation level model.

In order to accomplish batch level modeling the three-way batch data table is unfolded preserving batch direction. These models use the initial conditions plus summaries of the batch evolution as variables.

When batch level modeling is used for batch monitoring, the resulting PLS-model can be used to classify new evolving batches as either representative or non-representative. Another important objective is to understand how $\mathbf{Y}$ is influenced by the combination of initial conditions and batch evolution as well as aiding early fault detection. It is also possible to interpret which initial condition data and process evolution data exert the highest influence on the type and attributes of the resulting product.

Each batch is summarized by a set of batch level score values. Batches that cluster close to each other in the score space are similar and those far apart are different. For example shifts in raw material quality or step changes in media formulation are summarized as shifts or clusters in the score space. Similar to

the observation level contribution charts are used as a diagnostic tool to identify which variables are responsible for shifts.

Use Of MVDA for Analysis Of Cell Culture Data. MVDA is a multistep technique used to identify clusters, outliers and trends evident in cell culture process data, permitting the subsequent identification of correlations among key variables (Kirdar et al., 2007). The flowchart illustrated in Figure 2 presents an overview of the batch modeling when working with the SIMCA P+ software. The data must be imported into the program from the spreadsheet and an observation level model is fitted to the data through SIMCA-P+ software. Preparing representative data prior to fitting a representative model is an important step in the batch modeling process. The data must be pre-processed to identify outliers (data points strongly deviating from normal process behavior) and account for missing values. This is largely a manual process and requires a great deal of familiarity with the source data set as well as expertise on the process itself. The scientist/engineer can then review the models using several diagnostic plots and tools. Outliers with assignable root causes can be removed from the data set. In this step, it is crucial to remove the outliers but not the data within the process variability. If necessary, outlier batches or individual observations can be eliminated and a new model should be fit to the remaining data. The resulting model should be an appropriate fit to the data set and can be used for further statistical analysis, control charting or report generation.

SIMCA P+ generates many diagnostic plots that can be used to trend batch evolution as well as allow a greater understanding of the variable interactions driving the process. Table 1 presents the relevant MVDA diagnostic plots that are available from the analysis.

For the root cause application discussed, a large data set (171 runs) was available for each cell culture run, including continuous on-line measurements of operating parameters (pH, dissolved oxygen and temperature), daily measurements of dissolved CO₂, metabolic indicators and cell growth parameters. This data includes 2 L bench-scale, pilot-scale (2,000 L, 14 batches) and commercial-scale (15,000 L, 5 batches) data. A total of 119 output variables from raw materials, product attributes, time course and seed inocula trains were evaluated using MVDA.

Results and Discussion

Figure 3 shows an MVDA scores plot extracted from batch level model summarizing the output variables in large-scale data at 2,000 L (pilot, clinical) and 15,000 L (commercial). A visual summary of the process behavior over time can be seen in the scores plot, where the score vectors for the two principal components, $t[1]$ and $t[3]$, are plotted against each other. For our case, the principal component, $t[2]$, was summarizing variability and hence a $t[1]$ vs $t[3]$ plot was of greater interest to us. It is important to note that the percentage variability explained by the principal components when interpreting the significance of the plot (Kirdar et al., 2007). It is common to also plot an ellipse on this set of axis to represent the Hotelling T^2 95% confidence interval. This plot revealed differences between two data clusters shown as Group 1 and Group 2 in Figure 3. This plot differentiates process data on scale, culture performance and product attributes. In this plot, the x-axis shows score 1 (t_1) and the y-axis shows score 3 (t_3) where t_1 and t_3 are latent variables summarizing more than 100 variables in the data set. In this analysis, the variables that dominate the projection were raw materials, culture metabolism, growth

Table 1. Overview of MVDA Diagnostic Plots

Plot	Interpretation	Example	Theory
Scores plots	Overall batch evolution trends. Determine groups, trends, outliers and similarity within the workset. Elliptical confidence interval based on Hotelling T^2 . No anomalies should be seen in most influential scores plots.	t[1]/t[2]	Windows in the X -space displaying the observations as situated on the hyperplane.
		u[1]/u[2]	Windows in the Y -space displaying the observations as situated on the hyperplane.
		u[1]/t[1]	Display observations in the projected X (T) and Y (U) space. Shows how well the Y -space correlates with the X -space.
Loadings plots	Determine variables exerting most influence on batch evolution. Complements the respective scores plot.	w*1/ w*2	X * weights
		c1/ c2 w*c[1]/ w*c[2]	Y weights Shows both X and Y weights. Determines how the X and Y variables combine in the projections, and how the X variables relate to Y .
Contribution plots	Indicate which variables are the greatest contributors to a given process shift. Good starting point for fault analysis.	“Point and Click” (or use Contribution Tool) on any other plot to see variable contribution	Examines variable interactions and influence on the batch model at any given point during batch evolution.
Batch control charts	Detect outlying observations, process upsets.	Num/t[1]	Since the most variation is modeled by the first few components (contribute most to R^2 and Q^2), no anomalies should be seen in the first two to three scores.
Variable control charts	Use as single variable evolution/trace reference	VCD vs Run Time	± 3 SD and model average shown by default.
Distance to model plots	Shows unmodeled variation. A representative batch should evolve within the critical limits. Shows how the process points develop as a function of time, with easy identification of outliers.	DmodX.Comp1 or DmodY.Comp1	Used to compute the critical distance to the model, for observations in the work set, with the desired probability level. Observations outside the critical limits are outliers.
Variable importance for the projection (VIP) plot	Summarizes the observations made from the score and loading plots by showing the relative importance of each included variable in the analysis.	VIP(1) vs Variable ID	VIP values reflect the importance of the terms in the model both with respect to Y , i.e., its correlation to all the responses, and with respect to X (the projection).
Hotelling’s T^2 range plot	Summarizes the scores from second to the last component. T^2 plot is a combination of all the scores (t) in all components	T^2 Range	A measure of how far away an observation is from the center of the PCA or PLS model.
Column/line plot of loadings	Used to investigate the loadings for each score. Use the PLS Loadings plot p1 to interpret the scores t1, loadings p2 to interpret t2 etc.	p (1) vs Variable ID	Loadings p1 are the weights that combine the variables to form the scores t1. Score t1 is mainly a weighted combination of those variables that have a large loading in p1. Same for t2, t3, etc.

dynamics and product attributes. Group 1 contains commercial-scale runs performed at 15,000 L and pilot runs (P10–12) that tested specific raw material lots at 2,000 L scale. Group 1 data variables represent atypical product attributes, raw material type (basal medium powder), elevated culture lactate, elevated sodium (indicative of base addition) and elevated osmolality (atypical metabolism). This analysis directly correlated cell culture growth dynamics with metabolism and product attributes. The bioreactor process and product attributes are susceptible to a “cascade effect” of accumulating lactate, increased base addition and resulting high osmolality. In addition, the analysis demonstrates that the process is sensitive to raw material type. The effects of lactate metabolism and osmolality on cell culture growth and production have been reported (Zhu et al., 2005; DeZengotita et al. 1998; Ozturk et al., 1992).

SIMCA P+ generates several diagnostic plots that can be used to trend batch evolution as well as allow a greater understanding of the variable interactions influencing the process. To interpret the difference between two data clusters

in Figure 3, VIP, loading and contribution plots were also applied. Figure 4 illustrates the VIP plot focusing on variables that have the most influence on the process. This plot summarizes the observations made from the score and loading plots by showing the relative importance of each included variable in the analysis. The x-axis shows the primary variables that are important for the projection and the y-axis shows VIP values for each variable. The VIP values reflect the importance of the terms in the model with respect to **Y** (correlation to all the responses) and with respect to **X** (the projection). VIP values larger than 1 indicate significance. The VIP plot in Figure 4 indicates that large-scale raw material type and culture metabolism evolution exert the greatest influence on product attributes. These two parameters were further examined for their impact and interactions. Multivariate techniques take into account the interactions among variables to overcome the problems associated with univariate statistical process control and significantly reduce the number of charts that the process operator must observe (Montague et al., 2000).

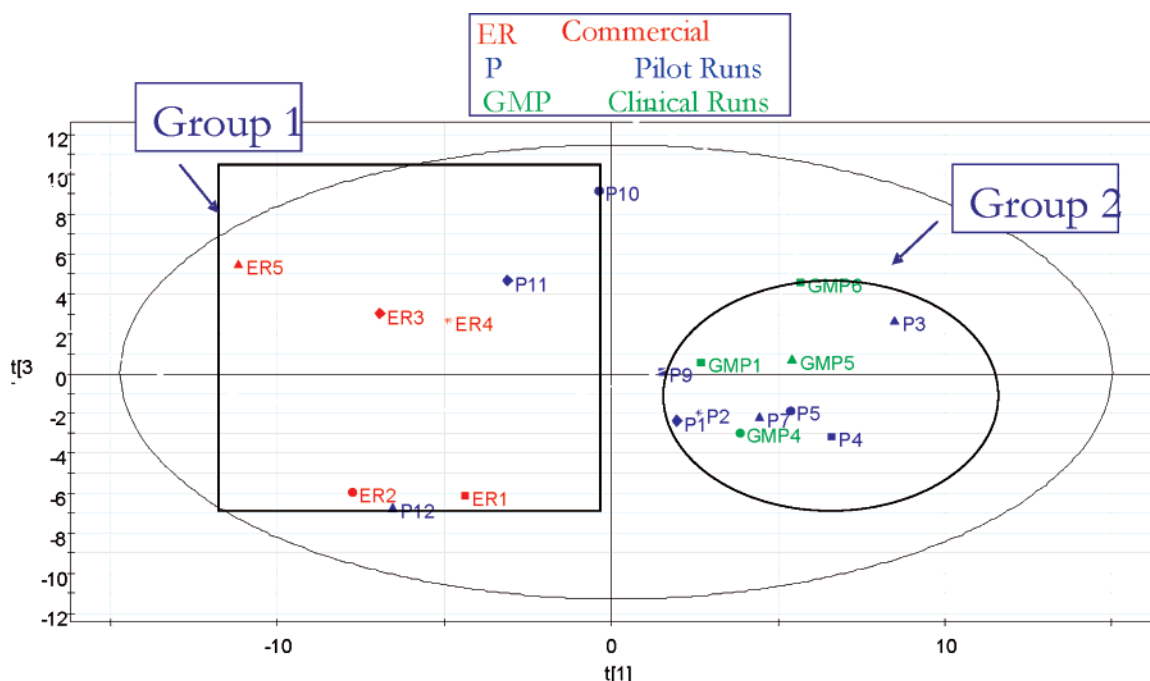


Figure 3. Score plot of process data related to scale, process performance and product attributes.

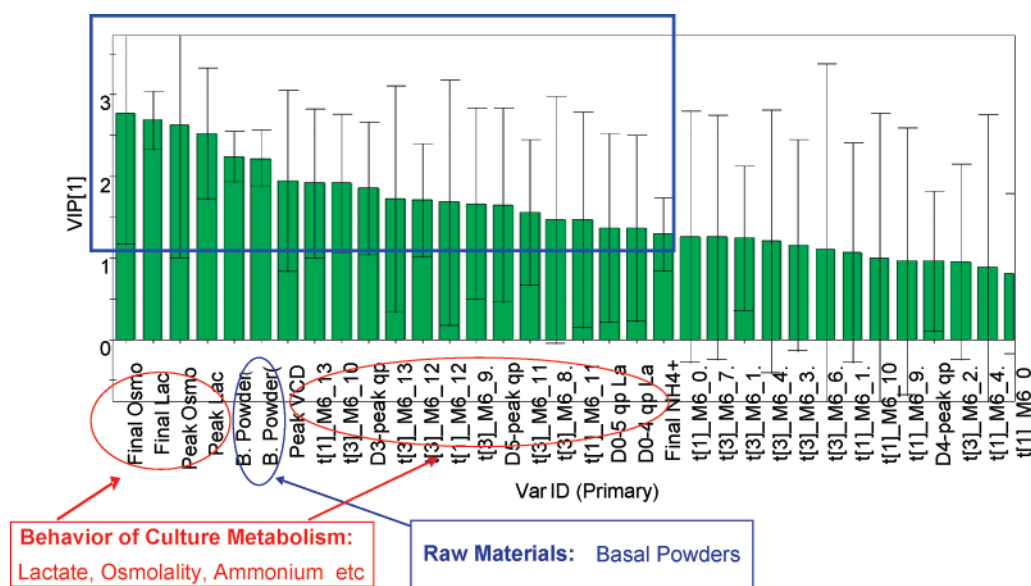


Figure 4. Variable importance for the projection (VIP) plot showing relative importance of all input and output variables included in the analysis.

Figure 5 illustrates a Variables Analysis Plot extracted from batch level model summarizing 2,000 L versus 15,000 L data sets focusing on product attributes with changes to raw material types. The x-axis shows the chronological order of runs (according to the date runs were performed) at pilot/clinical scale (2,000 L) and commercial scale (15,000 L) and the y-axis shows product attributes assay results. Evaluation of this data reveals a correlation between raw material type and product attributes. It is seen that there is a statistically relevant difference between process performance resulting from use of small-scale raw materials vs large-scale raw materials. During large-scale manufacturing of powder raw materials, limitations to conventional milling processes and the importance of performance qualification have been reported (Fike, 2001; Jayme et al., 2001; Radominski et al., 2001). These limitations include potential cross-contamination associated with dust generation, lengthy and incomplete processes of powder solubilization and dissolution,

the need to isolate “sensitive” constituents from the basal formulation and to add them separately to the hydrated powder, difficulty in achieving homogeneous distribution of minute formulation components and need for post-formulation pH and osmolality adjustment. (Jayme et al., 2001).

Figure 6 demonstrates results from coefficients plots created via MVDA batch level modeling of small-scale (2 L) and large-scale data (2,000 and 15,000 L). These plots confirmed the correlation of mid-culture bioreactor pCO₂ with the poor performing media type (small-scale media). The coefficients plot expresses how strongly product attributes (y-axis) are correlated to the systematic part of pCO₂ variable. The coefficient is considered significant or above noise level when the confidence interval does not cross zero. The x-axis is the days when pCO₂ was measured and the y-axis is the value of the regression coefficient (centered and scaled) for product attributes. A negative response for the plot on the right indicates high pCO₂

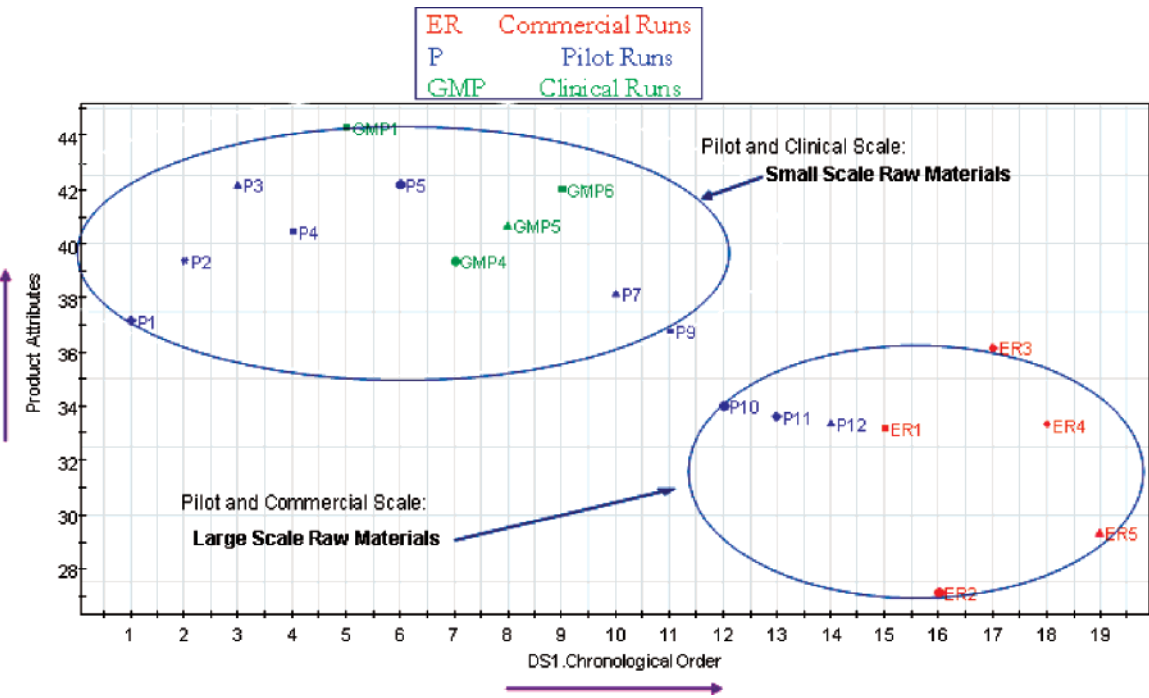


Figure 5. Variable analysis plot for 2,000 L and 15,000 L data sets.

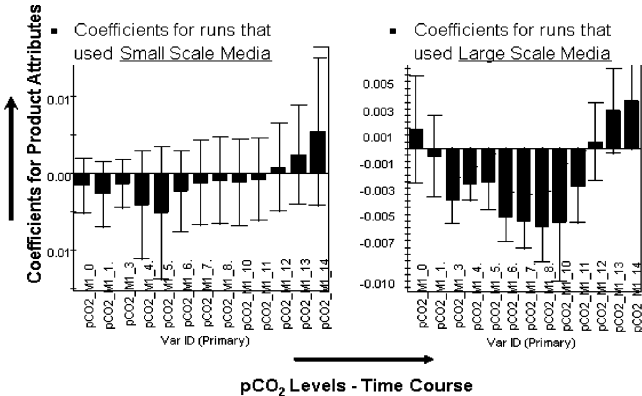


Figure 6. Coefficients plots using 2 L versus 2000 L, 15000 L data.

during this duration resulting in atypical product attributes. The data indicates that product attributes are more robust to elevated mid-culture pCO₂ levels when small-scale media is used. Process

issues associated with large-scale free suspension culture have become increasingly important as the scale increases (>10,000 L) in agitated bioreactors (Nienow et al., 2006). The detrimental effects of elevated bioreactor pCO₂ and osmolality on growth of CHO cells are well-documented in the literature (Gray et al., 1996; DeZengotita et al. 1998; Mostafa and Gu, 2003; Zhu et al., 2005).

MVDA results helped design the experimental work performed at 2,000 L pilot scale to successfully identify and correct the root cause with a limited number of pilot runs executed. Figure 7 summarizes the results from pilot runs performed at 2,000 L scale to analyze the effect of pCO₂ and basal media on the product attributes. Gas stripping to lower pCO₂ levels contributed to improved cell growth, metabolism and product attributes (using identical large-scale media), demonstrating process sensitivity to culture pCO₂. This sensitivity was alleviated with the reversion back to the original small-scale media and high pCO₂, indicating the interaction of elevated mid-culture pCO₂ and poor performing media (large-scale media).

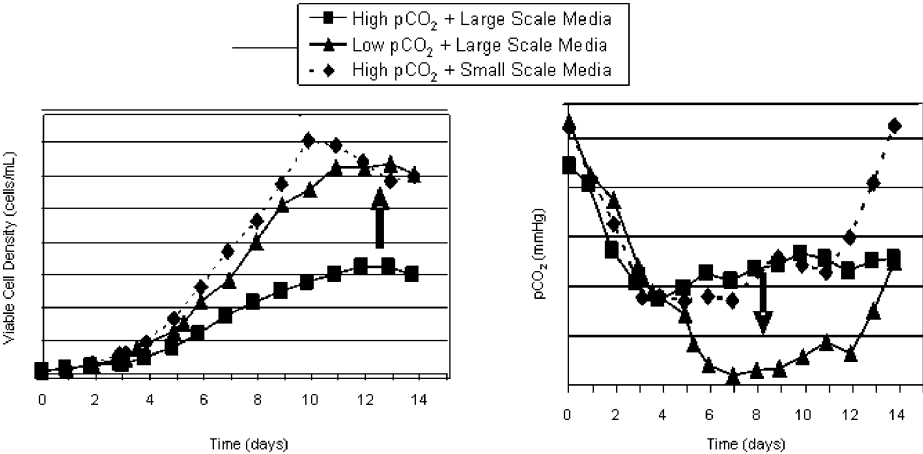


Figure 7. Confirmation of hypothesis via experiments at 2,000 L scale.

Conclusions

This paper demonstrates the diagnostic power of MVDA to extract useful process information through analysis of the readily available data in order to maximize process understanding. MVDA successfully identified the process parameters and their interactions that adversely impacted cell culture performance and product attributes and was able to predict culture performance trends across different scales. MVDA also helped design the experimental work, to successfully identify and correct the process root cause with only three pilot runs performed. This translated to time, resource and cost savings by focusing only on the relevant parameters.

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